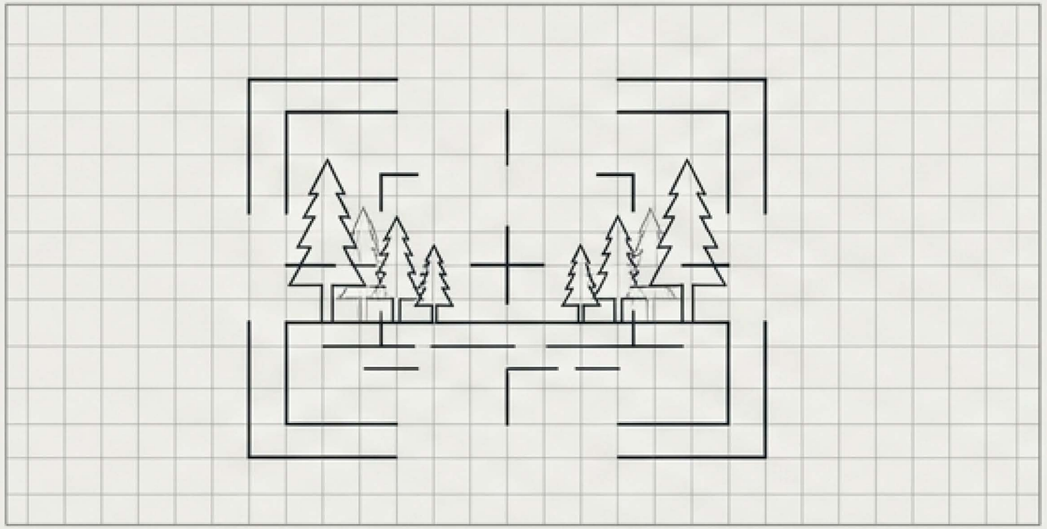
PARALLAX

Detecting what shouldn’t exist—even when it cannot be seen.

AI-based multi-sensor anomaly detection system.

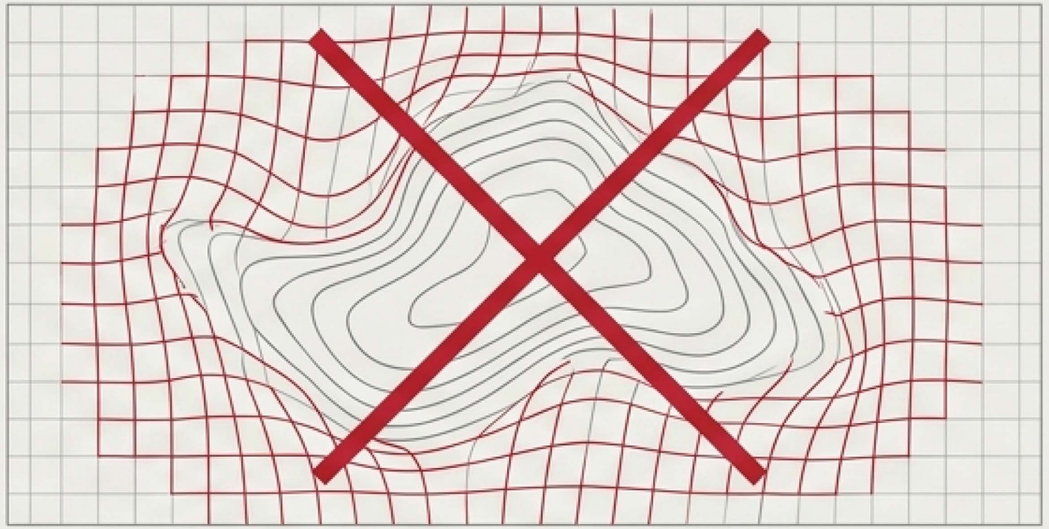
The Blind Spot in Modern Surveillance

Traditional Systems Rely On:



- Visual detection
- Thermal visibility

They Fail Against:



- Advanced camouflage
- Stealth materials
- Environmental blending

If you cannot see the object, current systems assume it does not exist.

Shifting to Anomaly Inference

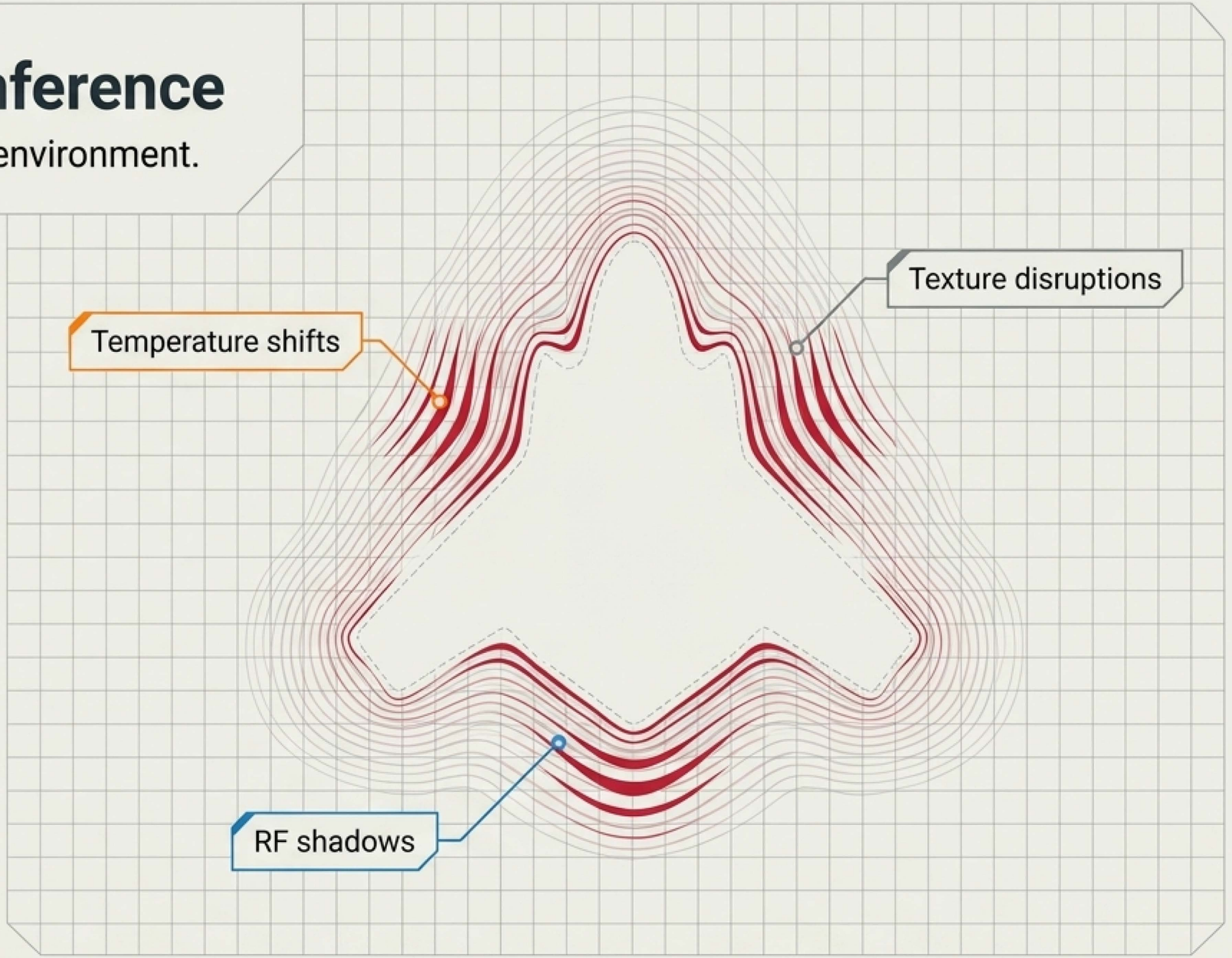
Hidden objects inevitably disturb their environment.

NOT Object Detection:

We do not rely on knowing what the threat looks like.

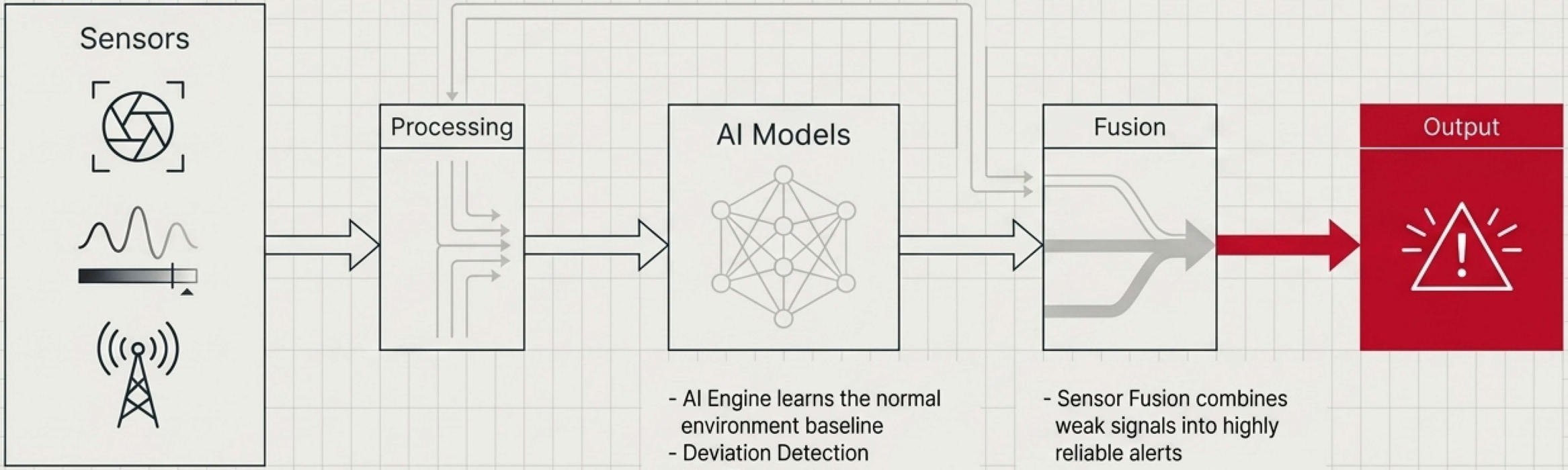
BUT Anomaly Detection:

We map the baseline environment and detect statistical deviations.



We detect what is out of place, not what is visible.

The Multi-Layered Detection Pipeline



Multi-sensor inputs:
- Vision (textures, motion)
- Thermal (heat inconsistencies)
- RF (signal disturbances)

- AI Engine learns the normal environment baseline
- Deviation Detection identifies statistical anomalies in real-time

- Sensor Fusion combines weak signals into highly reliable alerts

Deep-Tech Architecture

Autoencoders

Learns normal environmental patterns (High reconstruction error = anomaly).

Isolation Forest

Rapidly isolates rare and unusual data points.

The Multiplier Effect

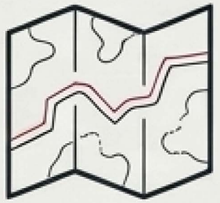
Combines weak, isolated signals (e.g., slight RF drop + minor heat shift) into a high-confidence detection.

Real-time Processing

Edge-optimized inference.

Built on Python, OpenCV, and TensorFlow / PyTorch

Scalable Security Beyond Visibility



Border Surveillance

Stopping undetected crossings.



Military Reconnaissance

Identifying camouflaged enemy assets.



Smart Security Systems

Critical infrastructure perimeter defense.

The PARALLAX Advantage

- Detects fundamentally hidden and unknown threats.
- Works beyond the limits of human and standard machine visibility.
- Highly scalable, multi-sensor reliability.